# CLIMATESHIELD MSME

Real-Time Climate Risk Scoring for MSMEs

**FUSION 2026 Hackathon — FIN-05**

A detailed problem analysis, data-source guide, solution architecture, scoring approach, team plan, rulebook checklist, presentation outline and winning strategy.

| | |
|---|---|
| **Problem owner / organization** | International Finance Corporation (IFC) |
| **Team size** | 4 students |
| **Core objective** | Translate climate exposure into explainable business and financial risk insights. |

## Guiding principle

**Do not build only a weather dashboard. Build a decision-support workflow that connects hazard evidence, business vulnerability, potential financial consequences and practical resilience actions.**

Prepared as a working planning document. Proposed formulas, thresholds and architecture are prototype recommendations—not official IFC standards or validated lending models.

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## How to use this guide

Use this document to align the team on scope, build a minimum viable product, validate data availability, and prepare a presentation. Keep a clear distinction between live data, downloaded data, user-entered values, simulated assumptions and features that are planned but not yet implemented.

# 1. Executive summary

FIN-05 asks teams to create a dynamic climate-adjusted financial risk score for micro, small and medium enterprises (MSMEs). Financial institutions may have similar historical financial information for two businesses but lack timely, granular evidence about differences in their exposure to floods, heatwaves, storms, drought or water shortages.

ClimateShield MSME is a proposed web platform that combines a business profile with relevant weather and climate information, geographic flood exposure and operational/financial vulnerability indicators. It returns an explainable 0–100 screening score, shows contributing factors and data confidence, simulates potential revenue disruption, and recommends preparedness actions.

## Recommended MVP

- Business registration/profile with coordinates or geocoded location, industry, revenue, cash buffer, utility arrangements and supplier dependencies.
- Climate hazard monitor using accessible weather forecasts and climate/flood datasets.
- Transparent rule-based score with factor breakdown and data-quality warnings.
- Financial impact simulator using clearly labelled hypothetical assumptions.
- Portfolio view comparing sample businesses and prioritizing cases for human review.
- Responsive dashboard with visible timestamps, API errors and fallback states.

## Core value proposition

**Make climate exposure visible before it becomes a financial crisis—while supporting business resilience and informed human review rather than automatically denying credit.**

The rulebook permits public APIs, datasets, open-source libraries and AI tools subject to the applicable rules and terms. It requires the team to build during the hackathon, understand the submitted work, provide required acknowledgements, and submit source code, a working prototype and presentation by the official deadline.

# 2. Deep analysis of the problem statement

Problem ID: FIN-05 — Real-Time Climate Risk Scoring for MSMEs. Organization named in the problem statement: International Finance Corporation (IFC).

## 2.1 Core problem

Small businesses in climate-exposed regions can suffer sudden revenue disruption after floods, heatwaves, storms or water shortages. A financial institution may not have the granular, timely information needed to distinguish two businesses with similar financial histories but very different physical and operational exposure.

## 2.2 What the expected solution implies

| Statement element | Implementation implication |
|---|---|
| **Real-time** | Refresh current/forecast-related indicators when available; show last-updated time and data source. |
| **Climate risk** | Separate immediate weather hazards from long-term climate vulnerability. |
| **Business location** | Use an address or coordinates to link the business to relevant geographic information. |
| **Satellite imagery** | Potentially add geographic imagery or hazard overlays if an accessible source and usable processing method are available. |
| **Flood/heat/drought** | Represent the relevant hazards with normalized indicators and source notes. |
| **Utility reliability** | Collect verified user-provided details or clearly labelled estimates if direct data are unavailable. |
| **Supply-chain exposure** | Record supplier/route dependencies and potential exposure to affected regions. |
| **Financial indicators** | Use available revenue, cash buffer, operating costs and debt obligations carefully. |
| **Lending and intervention** | Support assessment, monitoring, mitigation and human review; do not imply the score alone determines creditworthiness. |

## 2.3 The five questions the platform should answer

**1.** Where is the business, and which hazards are relevant to that location?

**2.** How severe are the current and forecast hazards, and what is the longer-term climate context?

**3.** How vulnerable are the business's operations, utilities and suppliers?

**4.** How might disruption affect revenue, cash flow and repayment pressure?

**5.** What practical actions can improve preparedness or reduce exposure?

## 2.4 Key distinction

**A climate hazard score is not the same as a probability of loan default. The proposed model is an indicative climate-related business-risk screening tool until it is validated against reliable business-loss and repayment outcomes.**

# 3. Target users and real-world examples

## 3.1 Target users

| User group | Potential need |
|---|---|
| **Banks, NBFCs and microfinance institutions** | Review climate exposure at application, renewal and portfolio-monitoring stages. |
| **MSME owners** | Understand vulnerabilities, prepare for disruptions and prioritize practical actions. |
| **Credit and risk analysts** | Compare cases consistently, inspect evidence and flag cases for human review. |
| **Insurers and business-support organizations** | Explore resilience planning, insurance assessment and targeted support applications. |

## 3.2 Illustrative comparison

Business A and Business B may have similar historical revenue. Business A has a lower-exposure location, cash reserves and alternative suppliers. Business B has greater flood exposure, depends on one access road and has weak backup power. A useful platform surfaces these differences, shows the evidence and explains how they may affect operations.

## 3.3 Related real-world examples

### Malaysia: flood risk and business resilience

World Bank research on flood risk and Malaysian businesses highlights the importance of SME resilience and supply-chain bottlenecks. Project lesson: include indirect disruption such as inaccessible roads and supplier delays.

Source: https://www.worldbank.org/en/country/malaysia/publication/flood-risk-management-leveraging-finance-for-business-resilience-in-malaysia

### Thailand: climate risk in banking

World Bank/Bank of Thailand-related work examined flood scenarios and geographic lending exposure. Project lesson: show lender portfolio concentration in vulnerable areas.

Source: https://blogs.worldbank.org/en/allaboutfinance/assessing-and-managing-natural-disaster-risks-for-the-financial-

### Kazakhstan: flood-affected entrepreneur

A World Bank account described a furniture workshop affected by 2024 floods and the difficulty of securing rebuilding credit. Project lesson: identify resilience and recovery support, not only risk.

Source: https://blogs.worldbank.org/en/psd/credit-guarantees-can-boost-business-resilience-in-central-asia

### India: climate finance

IFC announced a $500 million financing partnership with Axis Bank in October 2024 to expand green and blue finance. Project lesson: consider resilience investments as well as risk identification.

Source: https://www.ifc.org/en/pressroom/2024/ifc-partners-with-axis-bank-to-provide-a-500-million-climate-loan-in-india

These examples illustrate related problems and initiatives; they do not establish that the organizations use the proposed ClimateShield scoring formula.

# 4. Reference data platforms and limitations

The supplied Detailed Climate Websites Analysis PDF (pages 3–5) describes three complementary sources. Their capabilities are not interchangeable; each should be used only for indicators it actually supports.

| Source | What it offers | Proposed use / limitation |
|---|---|---|
| **World Bank Climate Change Knowledge Portal (CCKP)** | Historical climate indicators and future scenarios; temperature, precipitation, drought and heat-related indicators at regional/country scales. | Long-term climate context; avoid presenting broad regional indicators as precise property-level risk. Open reference |
| **Open-Meteo** | Weather forecasts, historical weather and related API services; structured JSON outputs. Availability depends on the selected API/model and its terms. | Near-term weather indicators for a business location. Check rate limits, parameters, attribution and data freshness. Open reference |
| **World Bank Global Flood Exposure Dataset** | The supplied PDF describes gridded exposure to a 1-in-100-year flood event and six flood-depth bands. | Geographic flood-exposure context. Population exposure is not equivalent to building/shop inundation or financial loss. Open reference |

## 4.1 Integration flow

**1.** Collect the business location and operational/financial profile.

**2.** Retrieve available weather forecasts and relevant climate/flood information.

**3.** Normalize each indicator and retain source, timestamp, units and confidence.

**4.** Combine available hazard, vulnerability, financial fragility, supplier and utility factors.

**5.** Return a score, factor breakdown, missing-data warnings and actionable guidance.

**6.** Store the score and scenario history for comparison and audit.

## 4.2 Important data limitations

- Long-term climate projections and short-term forecasts must be presented separately.
- Population exposure grids do not directly measure the flood probability of an individual business.
- Satellite imagery is mentioned in the problem statement but is not one of the three main platforms in the supplied reference PDF.
- Utility and supply-chain reliability may require user-entered information or labelled estimates.
- Test actual API/data availability early; a published portal does not guarantee that every desired indicator is available at the needed resolution.
- Credit all third-party datasets and comply with licenses and API terms.

# 5. Proposed solution: ClimateShield MSME

**Product positioning: real-time climate intelligence for smarter small-business lending and resilience.**

## 5.1 Core modules

| Module | Function |
|---|---|
| **MSME Climate Profile** | Business location, industry, revenue, cash reserves, utility arrangements and supplier dependencies. |
| **Climate Hazard Monitor** | Available current/forecast weather and geographic hazard indicators, with data timestamps. |
| **Explainable Risk Score** | 0–100 indicative screening score and factor-by-factor explanation. |
| **Financial Impact Simulator** | Hypothetical disruption days, revenue at risk and scenario comparisons. |
| **Early Warning and Actions** | Threshold-based alerts and practical preparedness suggestions. |
| **Lender Portfolio View** | Compare sample businesses, identify geographic concentration and prioritize human review. |

## 5.2 Suggested end-to-end user journey

**1.** Register a sample MSME and enter location and business indicators.

**2.** Retrieve available hazard information or explicitly load a labelled demo fallback.

**3.** Review hazard evidence and data confidence.

**4.** Calculate the indicative climate-related risk score.

**5.** Inspect the factors that most affect the score.

**6.** Run a hypothetical financial-impact scenario.

**7.** View practical preparedness actions and a lender-review summary.

## 5.3 Differentiation

Many dashboards show weather. ClimateShield's proposed differentiator is the connection between hazard, business vulnerability and financial consequence, followed by an intervention workflow. The score should not punish a business solely for being in a hazard-prone area; it should also reflect relevant resilience factors and data quality.

# 6. Proposed risk-scoring methodology

Use a transparent rule-based scoring model for the hackathon MVP. Avoid claiming that a predictive credit-risk model has been trained unless the team has suitable labelled historical data and has actually trained and evaluated it.

## 6.1 Example formula

$$R = 0.30H + 0.25V + 0.20F + 0.15S + 0.10U$$

All factors are normalized to a 0–100 scale. Higher values represent greater estimated climate-related business risk.

| Factor | Illustrative interpretation |
|---|---|
| **H — Hazard exposure (30%)** | Flood, heat, drought or relevant weather/climate indicators. |
| **V — Business vulnerability (25%)** | Physical/operational susceptibility, business type and preparedness. |
| **F — Financial fragility (20%)** | Cash buffer, revenue concentration and ability to withstand interruption, based on available inputs. |
| **S — Supply-chain exposure (15%)** | Supplier/transport dependency and disruption exposure. |
| **U — Utility exposure (10%)** | Power/water interruption dependence and backup arrangements. |

**The weights above are proposed prototype design choices. They are not IFC-approved, not supplied by the hackathon organizers, and not scientifically validated lending weights.**

## 6.2 Suggested score bands

| Score | Category | Suggested action |
|---|---|---|
| **0–24** | Low | Routine monitoring |
| **25–49** | Moderate | Review vulnerabilities and preparedness |
| **50–74** | High | Prioritize mitigation and lender review |
| **75–100** | Very high | Promptly assess exposure and intervention options |

These thresholds are illustrative and should not be presented as official lending standards.

## 6.3 Data confidence and missing values

- Display a separate confidence label based on data completeness, source quality, timestamp and geographic resolution.
- Do not silently replace missing hazard data with zero risk.
- Show whether each input is live API data, downloaded data, user-entered data or a simulation assumption.
- Record units, timestamps and source identifiers so a score can be explained later.
- Use human review for high-impact decisions; do not automatically approve or reject a loan based solely on this prototype score.

# 7. Financial impact simulator

The simulator converts a climate scenario into a simple business-impact estimate. All scenario assumptions must be visible and editable; the output is not a guaranteed forecast or a measured loss.

## Illustrative inputs

| Input | Example value |
|---|---|
| **Monthly revenue** | ■5,00,000 |
| **Operating days per month** | 25 |
| **Assumed disruption** | 6 days |
| **Revenue lost during disruption** | 80% |

## Gross revenue at risk = monthly revenue × (disruption days ÷ operating days) × assumed revenue-loss fraction

■5,00,000 × (6 ÷ 25) × 0.80 = ■96,000

This is a hypothetical estimate. Actual loss depends on inventory, fixed costs, insurance, recovery time, demand, alternative production and other factors.

## Scenario comparison

If a hypothetical preparedness plan reduced disruption from six days to two, keeping all other assumptions unchanged, the same simplified calculation would show ■32,000 of gross revenue at risk. This demonstrates scenario sensitivity only; it does not prove that the intervention will reduce actual downtime.

## Recommended visual output

- Baseline vs. disruption scenario vs. mitigation scenario.
- Estimated revenue at risk and assumptions shown beside the result.
- Top contributors to risk and recommended actions.
- Clear disclaimer that the simulator is an educational decision-support estimate.

# 8. Technical stack and architecture

Recommended architecture: React frontend → FastAPI backend → scoring/simulation modules → external data APIs and local database.

| Layer | Technology | Purpose |
|---|---|---|
| **Frontend** | React + Vite + TypeScript | Business form, dashboard and user workflow. |
| **Styling** | Tailwind CSS | Fast and consistent interface implementation. |
| **Charts** | Recharts | Risk factors and scenario comparisons. |
| **Mapping** | Leaflet + suitable map tiles | Location visualization; comply with tile-provider terms. |
| **Backend** | Python + FastAPI | Endpoints, data orchestration, score and simulator. |
| **Data processing** | Pandas + NumPy | Normalize and calculate indicators. |
| **Database** | SQLite | Sample business profiles, score history and scenario history. |
| **External data** | Open-Meteo, World Bank CCKP and flood dataset | Use only accessible and appropriate indicators. |
| **Testing** | Pytest + frontend tests | Scoring, API errors and user flow validation. |
| **Version control** | Git | Branches, small commits, backup and integration. |

## 8.1 Suggested API endpoints

| Endpoint | Purpose |
|---|---|
| **POST /businesses** | Create a business profile. |
| **GET /businesses/{id}** | Retrieve a profile. |
| **GET /businesses/{id}/climate** | Retrieve relevant hazard information. |
| **POST /businesses/{id}/score** | Calculate an updated score. |
| **POST /businesses/{id}/simulate** | Calculate a hypothetical financial-impact scenario. |
| **GET /portfolio/summary** | Summarize a sample portfolio. |

## 8.2 Error handling

- Set request timeouts and handle unavailable external APIs.
- Cache data when appropriate and show the retrieval timestamp.
- Label fallback/demo values prominently; never represent them as live data.
- Validate coordinate ranges, units and missing values.
- Keep API keys out of source control and use environment variables when secrets are needed.
- Return understandable error messages without exposing secrets or sensitive internal details.

# 9. AI-assisted development workflow

Five tools are listed by the team: Kilo CLI, OpenCode, Kiro, Antigravity and OpenAI Codex. Using all five on the same task can duplicate work and create conflicting edits. Give each a primary responsibility and verify capabilities in the actual installed setup.

| Tool | Suggested role |
|---|---|
| **Kiro** | Requirements, user stories, acceptance criteria and task breakdown. |
| **OpenCode** | Coding assistance, repository inspection and bounded implementation tasks, subject to configured models and permissions. |
| **OpenAI Codex** | Implementation tasks, unit tests, code review and bug fixing. |
| **Antigravity** | UI prototyping and end-to-end workflow assistance if the relevant capabilities are available. |
| **Kilo CLI** | Focused code changes, repetitive tasks and test assistance. |

## Operating rules

- Choose one primary coding environment; assign the other tools distinct tasks.
- Agree on API contracts, file ownership and branch strategy before parallel work.
- Do not let multiple agents edit the same files at the same time.
- Review and test generated code; every team member should be able to explain their work.
- Follow tool terms and any additional problem-specific restrictions.
- Do not prepare a pre-built solution before the hackathon; keep project coding, design and development within the permitted event period.

# 10. Four-member work division

## Member 1 — Product, team lead and integration

- Translate the problem statement into acceptance criteria and prioritize the MVP.
- Coordinate the scoring methodology and API contracts.
- Integrate modules, manage the feature checklist and maintain backups.
- Lead the pitch and coordinate the PPT and submission.

## Member 2 — Frontend and user experience

- Build the dashboard, business registration form and risk display.
- Implement factor breakdowns, charts and scenario comparison.
- Display data timestamps, missing-data warnings and recommendations.
- Test the complete user journey.

## Member 3 — Backend and risk engine

- Implement FastAPI endpoints and database models.
- Build the rule-based scoring function and factor breakdown.
- Implement the financial impact simulator.
- Write unit tests and validate sample profiles.

## Member 4 — Data integration and validation

- Test Open-Meteo endpoints and identify accessible climate/flood data.
- Normalize indicators and preserve units, timestamps and source details.
- Implement caching, timeouts, error handling and labelled fallback data.
- Validate calculations and prepare technical evidence for the demo.

**Shared responsibility: every member should understand the full system, test the integrated application and be ready to explain their contribution to judges.**

# 11. FUSION 2026 rulebook compliance

This checklist summarizes the supplied Timeline And Rulebook.pdf. The official document and any organizer updates take precedence.

| Requirement | Compliance action |
|---|---|
| **Team size and membership** | Team size must be 2–4 members; no changes after registration. |
| **Build period** | All coding, designing and development must be done during the hackathon; no pre-built work. |
| **Third-party resources** | Open-source libraries, public APIs and datasets are allowed subject to legal use, licenses, terms and appropriate credits. |
| **AI tools** | Allowed unless restricted by the problem statement. Team must understand code; required disclosure must not be misrepresented; excessive dependence may affect evaluation. |
| **Equipment** | Bring own laptops, chargers, extension board/power strip, cables/adapters and any required hardware. |
| **Backups** | Keep backups of code, files, presentation and important project data. |
| **Attendance and ID** | Follow physical verification and reporting instructions; carry original government/college ID as specified. |
| **Conduct** | No plagiarism, cheating, unauthorized access, impersonation, false registration or misconduct. |
| **Submission** | Submit source code, working demo/prototype and presentation through the official platform before deadline. |
| **Organizer authority** | Follow announcements and organizer instructions; violations can lead to disqualification. |

## 11.1 Published timeline in the supplied PDF

| Date/time shown in PDF | Milestone |
|---|---|
| **9 Oct, 6:00 AM** | New problem statements released |
| **9 Oct, 7:00 AM** | Venue check-in |
| **9 Oct, 10:00 AM** | Inauguration |
| **9 Oct, 2:00 PM** | Evaluation Round 1 |
| **9 Oct, 8:00 PM** | Evaluation Round 2 |
| **10 Oct, 11:00 AM** | Top 30 selection |
| **10 Oct, 4:00 PM** | Prize distribution |

These are the timeline entries printed in the uploaded rulebook, not confirmation of live changes. Verify the current submission cutoff and presentation duration with the organizers.

# 12. Judging criteria and winning strategy

The rulebook lists innovation, technical implementation, working prototype, problem–solution fit, feasibility, scalability, user experience, real-world impact, presentation/demo and overall execution. The uploaded document does not give numerical weights for these criteria.

## 12.1 Winning priorities

**1.** Deliver one complete and reliable workflow instead of many unfinished features.

**2.** Demonstrate two businesses with similar financial histories but different climate exposure and resilience.

**3.** Make the score explainable: factors, evidence, timestamps, missing information and limitations.

**4.** Make the financial impact simulator the memorable feature.

**5.** Prepare for technical questions on data provenance, scoring, API failure handling and limitations.

**6.** Finish with tested software, clean source code, backups, a concise presentation and a rehearsed demonstration.

## 12.2 What not to claim

- Do not claim the score predicts loan default unless that has been validated.
- Do not call a population exposure grid an exact shop-level flood probability.
- Do not claim simulated inputs are live data.
- Do not imply the suggested weights or thresholds are official IFC standards.
- Do not claim planned features are implemented; show only what works.

## 12.3 The memorable differentiator

Hazard → Business vulnerability → Financial consequence → Practical intervention.

A judge should see not only a risk number, but also why it was produced and what a business or lender can do next.

# 13. Proposed presentation outline

**The official PPT template has not been supplied in this conversation. This is a provisional content outline only; adapt the final slide order, count and formatting to the official template once available.**

### 1. Title and team

ClimateShield MSME; FIN-05; team name and four registered members.

### 2. Problem

Climate disruption creates differences that financial history alone may not reveal.

### 3. Solution

Business profile, hazard monitor, explainable score, impact simulator and actions.

### 4. Data and architecture

Open-Meteo, World Bank CCKP, flood exposure dataset and the processing workflow.

### 5. Scoring methodology

Proposed formula, factor weights, bands and confidence indicators.

### 6. Financial impact case

Sample business, disruption assumptions, revenue at risk and scenario comparison.

### 7. Technology stack

React, FastAPI, data connectors, SQLite, tests and Git.

### 8. Innovation and impact

Connect hazards to business consequences and resilience actions.

### 9. Feasibility and roadmap

Completed MVP, limitations, future features and scaling path.

### 10. Closing/demo

Core value proposition and transition to live demonstration.

## Presentation design principles

- Use real screenshots from the implemented application.
- Keep each slide focused on one main message and use readable labels.
- Use consistent color coding for risk categories without relying on color alone.
- Put source credits near data claims and include a compact reference slide if the template allows.
- Clearly distinguish implemented features, planned work and simulated values.

# 14. Pitch script and demo plan

Suggested three-minute pitch; adjust to the actual time limit announced by organizers.

## 0:00–0:30 — Problem

Two small businesses can have similar financial histories, but a flood or heatwave may affect their ability to operate very differently. Climate exposure is often missing from timely, granular business-risk assessments.

## 0:30–1:00 — Solution

ClimateShield MSME combines business information with climate and weather data to produce an explainable screening score, highlight vulnerabilities and estimate potential financial disruption.

## 1:00–2:30 — Demo

Select a sample business; show location and hazard data; inspect the score and its factors; compare a second business; run a disruption scenario; show practical preparedness actions.

## 2:30–3:00 — Impact

Our aim is to support better-informed lender review while helping MSMEs identify practical ways to prepare for climate disruptions.

## Live demo sequence

**1.** Open the dashboard and select or register a sample MSME.

**2.** Show the data source and last-updated time.

**3.** Generate and explain the risk score.

**4.** Compare the profile with a second sample business.

**5.** Run the financial impact simulator and change one assumption.

**6.** Show mitigation suggestions and the portfolio summary.

**7.** State one limitation honestly and explain how it would be validated in a real deployment.

## Fallback plan

- Have a clearly labelled demo dataset if an external API fails.
- Test the application offline or with cached responses before judging.
- Keep a short screen recording as a backup if permitted, but be ready to demonstrate the working prototype.
- Do not describe a fallback as live data.

# 15. Implementation roadmap and contingency plan

Suggested relative schedule for a 24-hour event. Adjust it to the official start, submission cutoff and judging slots.

## Phase 1 — Align and scaffold

Agree on MVP, score inputs, API contract, file ownership and repository; test one real API request.

## Phase 2 — Build the vertical slice

Implement profile input, one data connector, scoring endpoint and a simple score display.

## Phase 3 — Complete the demo workflow

Add factor breakdown, financial impact simulator, second-business comparison and portfolio summary.

## Phase 4 — Validate and harden

Test missing data, invalid inputs, API failure, score bounds, units and timestamp display.

## Phase 5 — Prepare deliverables

Capture screenshots, finalize slide content, rehearse the pitch, clean source code and verify backups/submission.

## Scope control: MVP first, enhancements second

- MVP: profile, weather/hazard data, transparent score, explanation and simulator.
- Next if time permits: portfolio map, multiple hazard types, report export and alerts.
- Stretch only if core flow is stable: satellite imagery analysis, richer supplier networks, scenario calibration and model-based predictions.

## If a data source fails

- Keep the source adapter isolated so the rest of the application remains usable.
- Use cached or synthetic demo values only with explicit labels.
- Explain the missing source and show which score components are affected.
- Never silently treat unavailable data as evidence of no risk.

# 16. API design, testing and final checklist

## 16.1 Minimum test cases

- Score stays within 0–100 for valid normalized inputs.
- Changing a factor changes the score in the expected direction.
- Missing inputs produce a warning or defined handling—not silent zero risk.
- An unavailable API produces a visible status and safe fallback.
- Financial simulator matches hand-calculated examples.
- Location, currency, units and timestamps are displayed correctly.
- Two businesses with different vulnerability inputs produce explainable differences.
- Frontend works from business entry to score and scenario result.
- Source citations and licenses are recorded.
- Fresh clone/restart instructions work on a team laptop.

## 16.2 Submission readiness checklist

- All four registered members know their roles and can explain their contribution.
- Project is built within the permitted hackathon period.
- All external resources and licenses are documented.
- API keys and secrets are not committed to source control.
- Working demo has been tested, including the fallback path.
- Source code is organized and runnable.
- Presentation follows the official template.
- Claims distinguish live data, downloaded data, user inputs and simulation assumptions.
- Score formula, weights, thresholds and limitations are documented.
- Code, presentation and important data have backups.
- Submission has been successfully uploaded before the official cutoff.
- Team has verified the final submission and monitored organizer announcements.

## 16.3 Suggested repository structure

```
climateshield-msme/
frontend/
src/
components/
pages/
services/
backend/
app/
main.py
api/
services/
weather_service.py
climate_service.py
risk_engine.py
impact_simulator.py
models/
tests/
data/
sample_businesses.json
docs/
data_sources.md
scoring_methodology.md
.env.example
README.md
```

# 17. References

## Primary materials supplied by the team

1. FIN-05 problem statement image — “Real-Time Climate Risk Scoring for MSMEs”; organization named: International Finance Corporation (IFC).

2. Timeline And Rulebook.pdf — FUSION 2026 rules, timeline, submission requirements and evaluation criteria.

3. Detailed_Climate_Websites_Analysis.pdf — summary of CCKP, Open-Meteo and the World Bank Global Flood Exposure dataset.

## Data sources named in the problem statement

4. World Bank Climate Change Knowledge Portal — Download Data
https://climateknowledgeportal.worldbank.org/download-data

5. Open-Meteo
https://open-meteo.com/

6. World Bank Global Flood Exposure dataset
https://datacatalog.worldbank.org/search/dataset/0062763/global-flood-exposure-gridded-exposure-headcounts-by-country

## Additional real-world reading used in the planning analysis

7. World Bank — Flood Risk Management: Leveraging Finance for Business Resilience in Malaysia
https://www.worldbank.org/en/country/malaysia/publication/flood-risk-management-leveraging-finance-for-business-resilience-in-malaysia

8. World Bank blog — Assessing and managing natural disaster risks for the financial sector
https://blogs.worldbank.org/en/allaboutfinance/assessing-and-managing-natural-disaster-risks-for-the-financial-

9. World Bank blog — Credit guarantees can boost business resilience in Central Asia
https://blogs.worldbank.org/en/psd/credit-guarantees-can-boost-business-resilience-in-central-asia

10. IFC — Partnership with Axis Bank to provide a $500 million climate loan in India
https://www.ifc.org/en/pressroom/2024/ifc-partners-with-axis-bank-to-provide-a-500-million-climate-loan-in-india

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## Final recommendation

**Win trust through a working, explainable and honest prototype. Use real data where possible, label assumptions, show a business-level consequence and make it clear how the result supports human decisions. Build a complete workflow before adding complexity.**

Prepared for planning and presentation development. Verify data access, current API terms, exact event deadlines and any organizer updates before submission.